

POSTS AND TELECOMMUNICATIONS INSTITUTE OF TECHNOLOGY
FACULTY OF INFORMATION TECHNOLOGY 1



FINAL REPORT

BUSINESS INTELLIGENCE

Topic: BI system for Movie data analysis

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TABLE OF CONTENTS

PREFACE	3
CHAPTER 1: INTRODUCTION	4
1. Background and analysis objectives.....	4
2. Scope of the Business Intelligence (BI) system	5
CHAPTER 2: DATASET DESCRIPTION	6
1. Data source and collection method.....	6
2. Details of key attributes	6
CHAPTER 3: DATA WAREHOUSE DESIGN	8
1. Analysis and identification of Fact and Dimension tables	8
1.1. Fact Table.....	8
1.2. Dimension Table	8
1.3. Bridge Table.....	9
2. Data model diagram.....	9
CHAPTER 4: ETL PROCESS.....	11
1. Processing tools	11
2. Description of ETL steps.....	12
CHAPTER 5: DASHBOARD DEVELOPMENT	16
1. Data visualization tool	16
2. Key charts and metrics.....	16
CHAPTER 6: INSIGHT ANALYSIS	20
1. Insights derived from data visualization.....	20
2. Trend evaluation and recommendations.....	24
CHAPTER 7: CONCLUSION	26
1. Results.....	26
2. Limitations and future development.....	26
REFERENCES	28

PREFACE

First, we would like to sincerely thank the Posts and Telecommunications Institute of Technology and the Faculty of Information Technology 1 for including the Business Intelligence course in the curriculum. We are especially grateful to our lecturer, Mr. Kim Ngoc Bach, for your dedication, guidance, and the valuable knowledge you have shared with us throughout the course.

Through your lessons, we have not only gained a deeper understanding of the subject but also developed a more disciplined and effective approach to our work. These experiences will undoubtedly support us in our future studies and careers. With your thoughtful guidance, we were also able to complete our term project successfully, for which we are truly thankful.

Once again, we would like to express our heartfelt appreciation and wish you good health and continued success in both your life and career.

CHAPTER 1: INTRODUCTION

1. Background and analysis objectives

Nowadays, with the explosion of the digital era and the internet, the film and online entertainment industry is witnessing outstanding development. Giant streaming platforms like Netflix, combined with large-scale database systems like IMDb, continuously provide millions of hours of cinematic content to global users, serving the increasingly diverse entertainment and cultural needs of humanity.

As the volume of content and data becomes massive, a core problem arises for producers, investors, and distribution platforms: how to truly understand audience tastes. Producing, directing marketing strategies, or purchasing the licensing rights of a film without grasping viewer trends can lead to significant revenue risks. Furthermore, critical information such as viewer ratings, trending genres, or the shifts in revenue streams over the years is often vaguely dispersed within raw datasets. Especially in the context of the rise of Data Science, making decisions based on personal intuition is no longer effective; instead, businesses need to thoroughly mine data to uncover valuable insights. Stemming from the digital transformation trend of the market and the abundant available data, our team recognizes this as a highly engaging analytical problem with high practical application and therefore decided to choose this direction as the main research topic for our major assignment.

To effectively solve the problem of cinematic data mining, our team decided to implement the project: BI system for Movie data analysis. This system will act as a complete processing pipeline to help collect, clean, and visualize historical data from reputable sources. Through multidimensional analytical dashboards, the system allows users to easily track the popularity of film genres, compare

revenue across different years, and conduct in-depth analysis of viewer rating behaviors. The project's product aims to provide a comprehensive overview, strongly supporting data-driven decision-making in the film industry.

2. Scope of the Business Intelligence (BI) system

The scope of this project focuses on the data processing and visualization pipeline for a historical dataset of 5000 movies, specifically encompassing:

- Data scope: Mining and analyzing core movie attributes, which include budget, revenue, viewer ratings, vote counts, genres, release dates, and popularity.
- Data processing (ETL) scope: Extracting and cleansing raw data and organizing it into a multi-dimensional data model comprising Fact and Dimension tables.
- Analytical and visualization scope: Developing interactive dashboards to deliver reports centered around three core themes:
 - Movie distribution and performance (revenue/profit) segmented by genre.
 - Trends in revenue and profit over the years.
 - Analysis of viewer rating behaviors combined with overall movie popularity.

CHAPTER 2: DATASET DESCRIPTION

1. Data source and collection method

- **Data source:** <https://www.kaggle.com/datasets/tmdb/tmdb-movie-metadata>

The project utilizes the “TMDB 5000 Movie Dataset” publicly available on the Kaggle platform. This dataset contains detailed information on approximately 5,000 movies, originally extracted from The Movie Database (TMDb). The raw dataset consists of two primary files:

- `tmdb_5000_movies.csv`: Contains the metadata attributes of the movies such as budget, revenue, release date, user ratings (vote average, vote count), genres, and popularity.
- `tmdb_5000_credits.csv`: Contains detailed information regarding the cast and production crew for each movie.

Note: in the current scope only `tmdb_5000_movies.csv` feeds the warehouse; `tmdb_5000_credits.csv` is read but not yet used downstream - it is reserved for a future cast/crew dimension.

- **Collection method:**

This static dataset has already been aggregated by the community via the TMDb API and published on Kaggle in .csv format. Therefore, the data was collected by directly downloading these files from the Kaggle website to local storage, completely bypassing the need for manual web scraping or custom API calls.

2. Details of key attributes

Although the original dataset contains a vast amount of information, to serve the BI system’s goals of analyzing trends, revenue, and ratings, the project

specifically extracts and utilizes only the core attributes that provide analytical value. Below are the most critical data fields:

- Identification and general information attributes:
 - id: A unique identifier for each movie within the TMDb system.
 - title/ original_title: The official name of the movie.
 - genres: A list of categories the movie belongs to (e.g., Action, Adventure, Sci-Fi). In the raw dataset, this is stored as a JSON-like string containing genre IDs and names.
 - release_date: The official theatrical release date, which serves as the foundation for creating time dimensions (Year, Quarter, Month) for trend analysis.
 - runtime: The total duration of the movie, measured in minutes.
- Financial attributes:
 - budget: The production cost of the movie (in USD).
 - revenue: The total worldwide box office gross of the movie (in USD).
- Derived attribute:
 - profit: an internally calculated field obtained by subtracting budget from revenue, used to evaluate the commercial success of the film.
- Rating and engagement attributes:
 - vote_average: The average rating score given by the viewer community (on a 10-point scale).
 - vote_count: The total number of user votes, which helps determine the statistical reliability of the average rating.
 - popularity: A specific metric calculated by TMDb that reflects the trending status and overall community interest in the movie.

CHAPTER 3: DATA WAREHOUSE DESIGN

1. Analysis and identification of Fact and Dimension tables

To construct an optimized data warehouse for querying and reporting movie performance, the data model is designed using a star schema, with an additional bridge table to resolve the many-to-many relationship between movies and genres. The analysis of business entities and quantitative metrics has led to the identification of the following Fact, Dimension, and Bridge tables:

1.1. Fact Table

Table FactMovies: This sits at the center of the data warehouse model, storing foreign keys that connect to the dimensions alongside quantitative, aggregable measurements (metrics/measures).

- Primary/Foreign Keys: MovieKey (links to the movie information dimension), DateKey (links to the time dimension).
- Measures: Budget, Revenue, Profit, Runtime, Popularity, VoteAverage, and VoteCount. These metrics directly support aggregation operations (such as SUM and AVERAGE) on the interactive dashboards.

1.2. Dimension Table

Dimension tables provide descriptive context, enabling users to filter, group, and slice the quantitative data stored within the Fact table.

- Table DimMovieInfo: Stores the identifying attributes and core textual metadata for each unique motion picture.

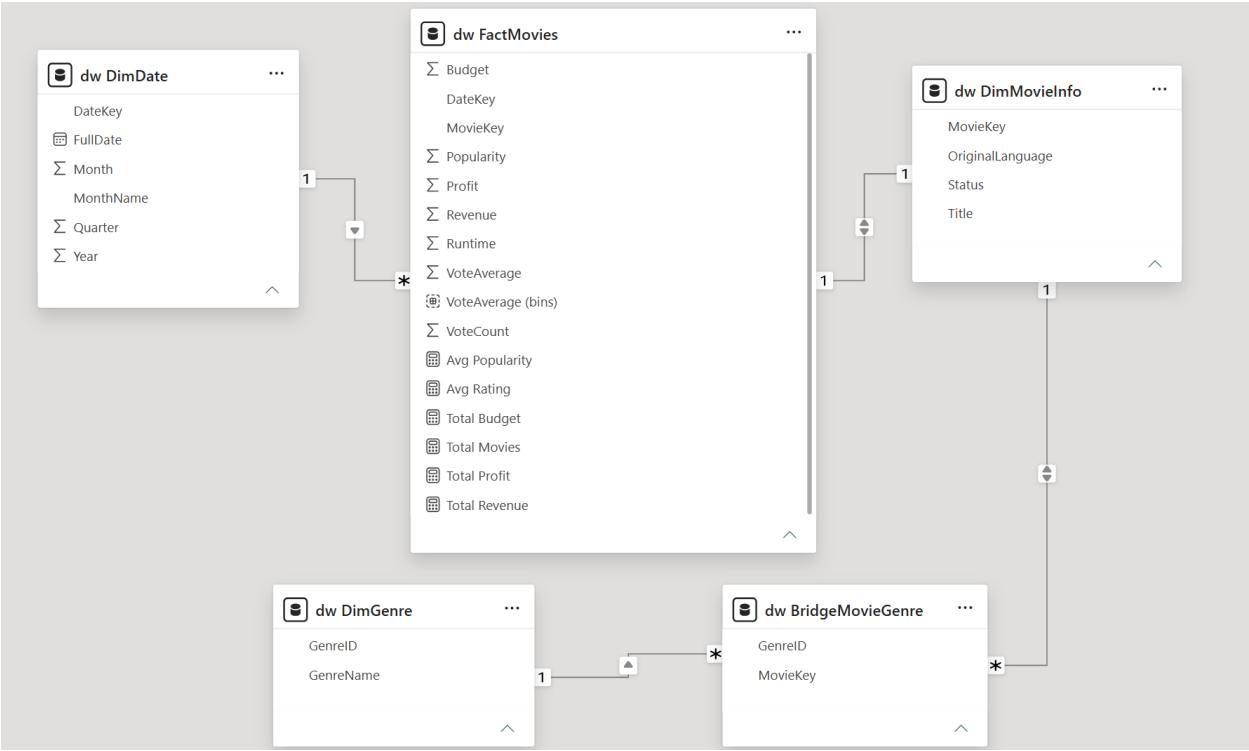
- Attributes: MovieKey (Primary Key), Title, OriginalLanguage, and Status. This dimension allows users to segment financial performance by specific titles or linguistic origins.
- Table DimDate: A dedicated time dimension designed to support historical trend analysis across various temporal granularities.
 - Attributes: DateKey (Primary Key), FullDate, Year, Quarter, Month, and MonthName. This table is essential for generating time-series line charts.
- Table DimGenre: The genre dimension houses the master list of movie classifications to facilitate audience preference analysis.
 - Attributes: GenreID (Primary Key) and GenreName (e.g., Action, Comedy, Thriller).

1.3. Bridge Table

Table BridgeMovieGenre: Since a single movie can belong to multiple genres and a single genre contains multiple movies, the relationship between DimMovieInfo and DimGenre is Many-to-Many. This bridge table is implemented to normalize the schema and prevent data inflation or duplicate entries inside the Fact table.

- Attributes: MovieKey and GenreID (comprising a composite primary key while acting as foreign keys linking the two respective dimensions).

2. Data model diagram



CHAPTER 4: ETL PROCESS

1. Processing tools

Based on the suggested tools in the assignment guidelines and the specific complexities of the dataset, the ETL (Extract - Transform - Load) pipeline is built using a code-centric approach rather than relying on drag-and-drop GUI ETL software (such as Pentaho or Talend). Specifically, the selected tools are:

- Extract & transform tool: Python
 - Description: The system utilizes Python within a Jupyter Notebook environment, heavily relying on the pandas data manipulation library, alongside built-in modules like json and ast.
 - Reason for selection: The raw dataset contains fields stored as complex JSON-like strings (notably the genres column). Python demonstrates maximum flexibility and superiority in parsing, cleansing erroneous values, and exploding these arrays into structured table formats compared to visual tools. Writing custom Python scripts allows for complete and optimal control over the data transformation logic.
- Load Tool & Database: Microsoft SQL Server (T-SQL)
 - Description: Microsoft SQL Server is employed as the Data Warehouse relational database management system, utilizing T-SQL scripts to execute the Load phase.
 - Reason for selection: While the syllabus suggests MySQL or PostgreSQL, the system leverages SQL Server to take advantage of the highly efficient BULK INSERT command. This command enables the rapid, high-volume loading of data from the Python-processed CSV files into intermediate staging tables. Subsequently, T-SQL scripts safely handle data type casting and the final insertion of data into the

official Fact and Dimension tables, guaranteeing the referential integrity of the entire Data Warehouse architecture.

2. Description of ETL steps

The ETL process is designed as a rigorous pipeline, divided into two primary phases: Pre-processing with Python (Extract & Transform) and loading data with T-SQL (Load).

Phase 1: Extract & Transform (Using Python)

- Extract: The system reads the raw data from the `tmdb_5000_movies.csv` file using the pandas library.
- Transform & Cleanse:
 - Type casting: Quantitative columns (e.g., `budget`, `revenue`, `vote_average`) and identifiers (`id`) are strictly cast to numeric types. The `release_date` column is converted into a Datetime format.

```
movies = data_movie.copy()
movies['MovieKey'] = pd.to_numeric(movies['id'], errors='coerce').astype('Int64')
movies['release_date'] = pd.to_datetime(movies['release_date'], errors='coerce')

numeric_cols = ['budget', 'revenue', 'vote_average', 'vote_count', 'popularity', 'runtime']
for col in numeric_cols:
    movies[col] = pd.to_numeric(movies[col], errors='coerce')

movies['title'] = movies['title'].fillna(movies['original_title'])
movies['original_language'] = movies['original_language'].apply(normalize_string)
movies['status'] = movies['status'].apply(normalize_string)
movies['genres_parsed'] = movies['genres'].apply(parse_json_like)
movies['DateKey'] = pd.to_numeric(movies['release_date'].dt.strftime('%Y%m%d'), errors='coerce').astype('Int64')
```

- Parsing complex data: The `genres` field (stored as a JSON-like string) is parsed using custom functions (utilizing `ast.literal_eval` and `json.loads`) to extract them into workable lists containing genre IDs and Names.

```
def parse_json_like(value):
    if pd.isna(value):
        return []
    if isinstance(value, list):
        return value
    if not isinstance(value, str) or not value.strip():
        return []
    try:
        parsed = ast.literal_eval(value)
        return parsed if isinstance(parsed, list) else []
    except (ValueError, SyntaxError):
        try:
            parsed = json.loads(value)
            return parsed if isinstance(parsed, list) else []
        except Exception:
            return []
```

- Handling missing values: Missing text strings are imputed with a default value (“Unknown”). The title column is filled using original_title if left blank.

```
def normalize_string(value, default='Unknown'):
    if pd.isna(value) or str(value).strip() == '':
        return default
    return str(value).strip()
```

- Handling placeholder zeros in financial fields: In TMDB, a budget or revenue of 0 denotes unknown, not an actual \$0 figure. These zero (and negative) values are therefore converted to NULL so that financial aggregations (SUM/AVG of Budget, Revenue) are not distorted. Consequently, Profit = Revenue - Budget is computed only when both Budget and Revenue are known; otherwise, it is left NULL. The movie rows themselves are kept, so genre counts and rating analysis still cover

the full catalogue (~4,803 movies); after filtering, ~3,376 movies have a valid Revenue, ~3,766 a valid Budget, and ~3,229 a valid Profit.

```
for col in ['Budget', 'Revenue']:
    fact_movies[col] = fact_movies[col].where(fact_movies[col] > 0)

fact_movies['Profit'] = fact_movies['Revenue'] - fact_movies['Budget']

for col in ['Budget', 'Revenue', 'VoteCount', 'Profit']:
    fact_movies[col] = fact_movies[col].astype('Int64')
```

- Structuring: The cleansed data is split and modeled into Dimension tables (DimDate, DimGenre, DimMovieInfo), the Fact table (FactMovies, which includes an additionally calculated Profit column = Revenue - Budget), and a Bridge table (BridgeMovieGenre created by splitting the genre lists into separate rows). These tables are subsequently exported as clean .csv files.

```
genre_expanded = movies[['MovieKey', 'genres_parsed']].explode('genres_parsed').dropna(subset=['genres_parsed']).reset_index(drop=True)
genre_details = pd.concat([genre_expanded[['MovieKey']], genre_expanded['genres_parsed'].apply(pd.Series)], axis=1)

dim_genre = (
    genre_details[['id', 'name']]
    .rename(columns={'id': 'GenreID', 'name': 'GenreName'})
    .drop_duplicates()
    .sort_values(['GenreID', 'GenreName'])
    .reset_index(drop=True)
)
```

Phase 2: Load (Using SQL Server)

- Staging: The BULK INSERT command is utilized to rapidly load the clean .csv files into staging tables within SQL Server. These staging tables use safe data types (primarily VARCHAR) to prevent pipeline failures caused by unexpected anomalous data.

```
/* BULK INSERT helper settings for UTF-8 CSV with header */
SET @SQL = N'BULK INSERT #stgDimDate FROM ''' + REPLACE(@BasePath + N'DimDate.csv', '\', '\\') + N''''
WITH (
    FORMAT = 'CSV',
    FIRSTROW = 2,
    FIELDQUOTE = '"""',
    CODEPAGE = '65001',
    TABLOCK
);
EXEC sp_executesql @SQL;
```

- Loading into the Data warehouse: Data from the staging tables is finally inserted into the normalized structural tables (e.g., dw.DimDate, dw.FactMovies). During this step, TRY_CONVERT functions are applied to perform a final data type validation, safely ignoring records with null Primary Keys, thereby ensuring the referential integrity of the Star schema model.

```
/* Load dimensions first */
INSERT INTO dw.DimDate (DateKey, FullDate, [Year], [Quarter], [Month], MonthName)
SELECT
    TRY_CONVERT(INT, NULLIF(LTRIM(RTRIM(DateKey)), '')),
    TRY_CONVERT(DATE, NULLIF(LTRIM(RTRIM(FullDate)), '')),
    TRY_CONVERT(SMALLINT, NULLIF(LTRIM(RTRIM([Year])), '')),
    TRY_CONVERT(TINYINT, NULLIF(LTRIM(RTRIM([Quarter])), '')),
    TRY_CONVERT(TINYINT, NULLIF(LTRIM(RTRIM([Month])), '')),
    COALESCE(NULLIF(LTRIM(RTRIM(MonthName)), ''), N'Unknown')
FROM #stgDimDate
WHERE TRY_CONVERT(INT, NULLIF(LTRIM(RTRIM(DateKey)), '')) IS NOT NULL;
```

CHAPTER 5: DASHBOARD DEVELOPMENT

1. Data visualization tool

In this project, the team selected Microsoft Power BI as the primary tool for data visualization and dashboard development.

- Description: Power BI is a robust business analytics platform by Microsoft. It enables users to connect to diverse data sources, build data models, and design interactive dashboards using an intuitive drag-and-drop interface combined with the DAX (Data Analysis Expressions) formula language.
- Reason for selection: Power BI offers seamless native integration with SQL Server. Its Model View is exceptionally well-suited for managing the Star Schema (specifically handling the many-to-many relationship via the BridgeMovieGenre table). Additionally, DAX allows for the dynamic calculation of Measures based on user filters, and its rich visual library perfectly fulfills all the core analytical requirements of the project.

2. Key charts and metrics

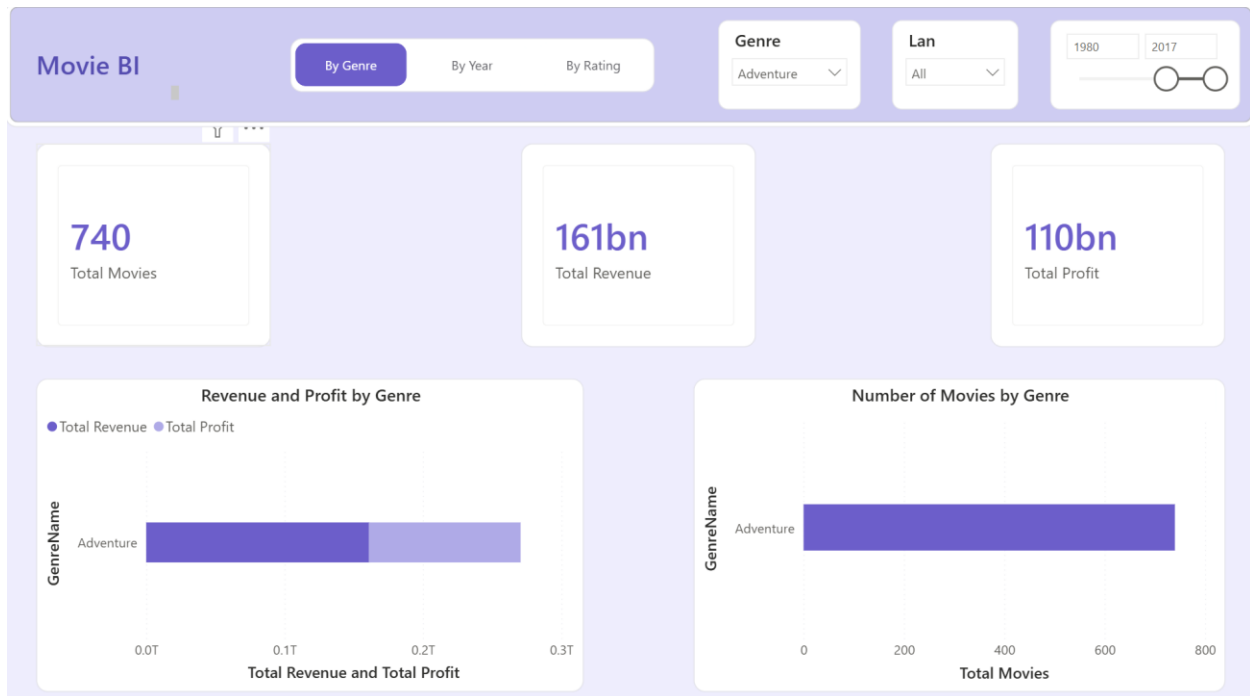
The dashboard is organized into three report pages, sharing a common header (title, page navigator) and three slicers - Genre, Original Language, and Release Year - so every view can be filtered consistently.

Core DAX measures: Total Movies (movie count), Total Revenue, Total Budget, Total Profit (SUM), Avg Rating (AVERAGE of VoteAverage), Avg Popularity (AVERAGE of Popularity).

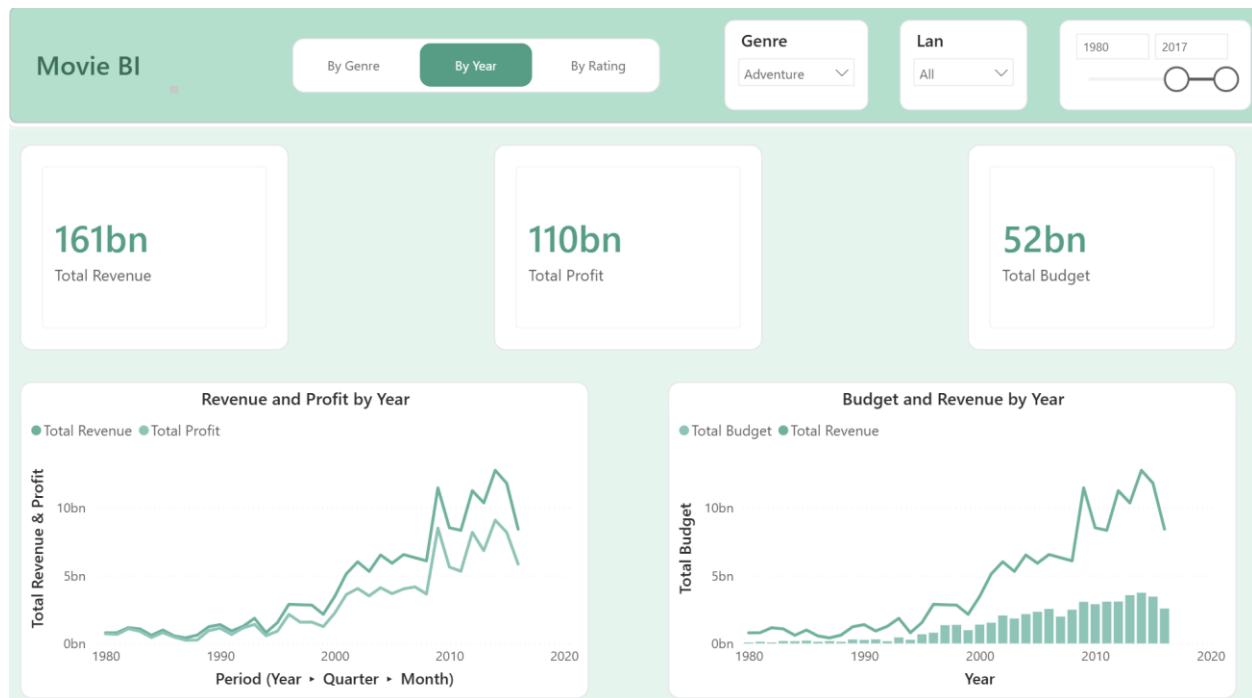
- By Genre: KPI cards (Total Movies, Total Revenue, Total Profit); a bar chart “Number of Movies by Genre” (GenreName x Total Movies); and a bar chart

“Revenue and Profit by Genre” (GenreName x Total Revenue & Total Profit).

Answers about which genres are most produced and most profitable.



- By Year: KPI cards (Total Revenue, Total Profit, Total Budget); a line chart “Revenue and Profit by Year” over a Year → Quarter → Month hierarchy; and a line-and-column combo chart “Budget and Revenue by Year”. Answers how the market’s financial performance trends over time.



- By Rating: KPI cards (Avg Rating, Avg Popularity, Total Movies); a scatter chart “Rating vs Popularity” (X = Avg Rating, Y = Avg Popularity, bubble size = Total Revenue, one point per Title); and a column chart “Rating Distribution” (VoteAverage bins x Total Movies). Answers how viewers rate movies and whether higher ratings relate to popularity/revenue.

Movie BI

By Genre

By Year

By Rating

Genre

Adventure

Language

All

1980

2017



6.12

Avg Rating

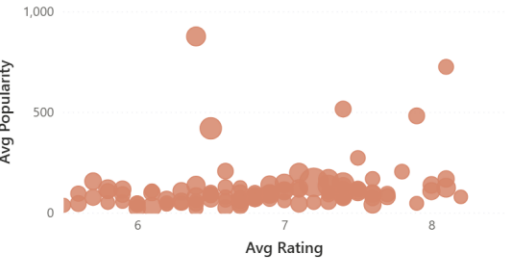
40.33

Avg Popularity

740

Total Movies

Rating vs Popularity



CHAPTER 6: INSIGHT ANALYSIS

1. Insights derived from data visualization

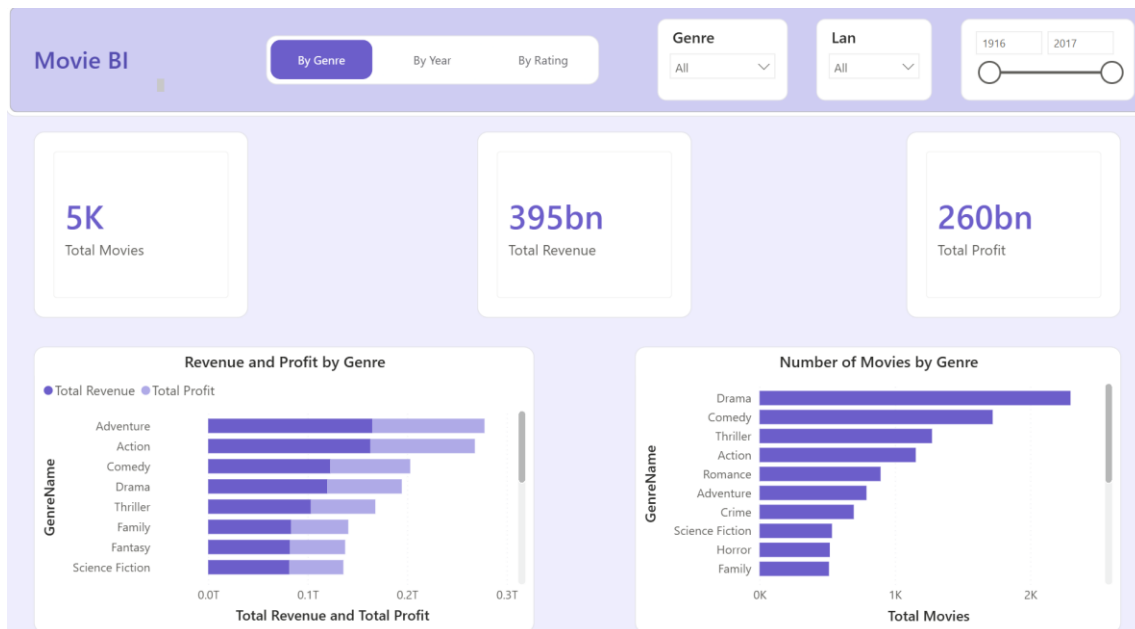
– Overview:

- ~5,000 films generate ~\$395bn in revenue and ~\$256bn in profit against a ~\$140bn budget.
- Since Profit = Revenue - Budget, the aggregate profit margin is ~ 65% and ROI on budget is ~1.8x.
- However, these overall figures are significantly inflated by a small group of blockbusters.

– Genre:

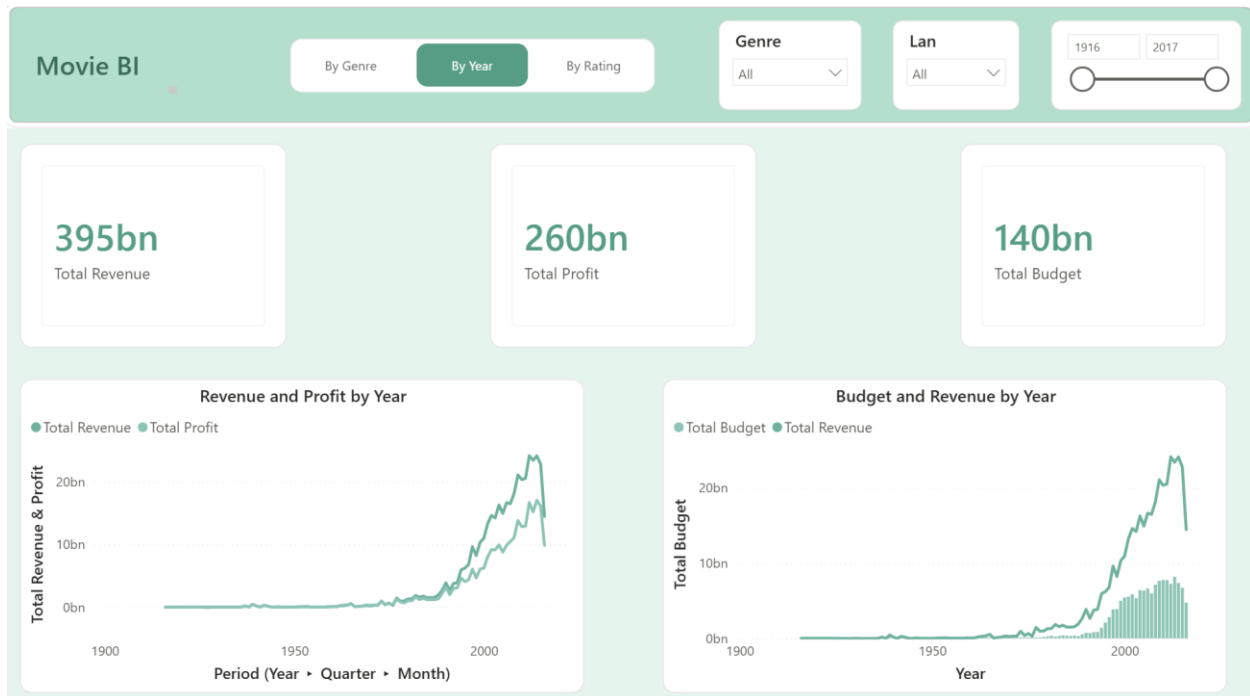
- Drama, Comedy, and Thriller are the most produced genres (Drama alone accounts for > 2,000 films).
- Adventure and Action generate the most revenue and profit despite having far fewer titles - meaning their revenue per film is much higher.

→ Evaluate genres by per-film performance, not just total production volume.



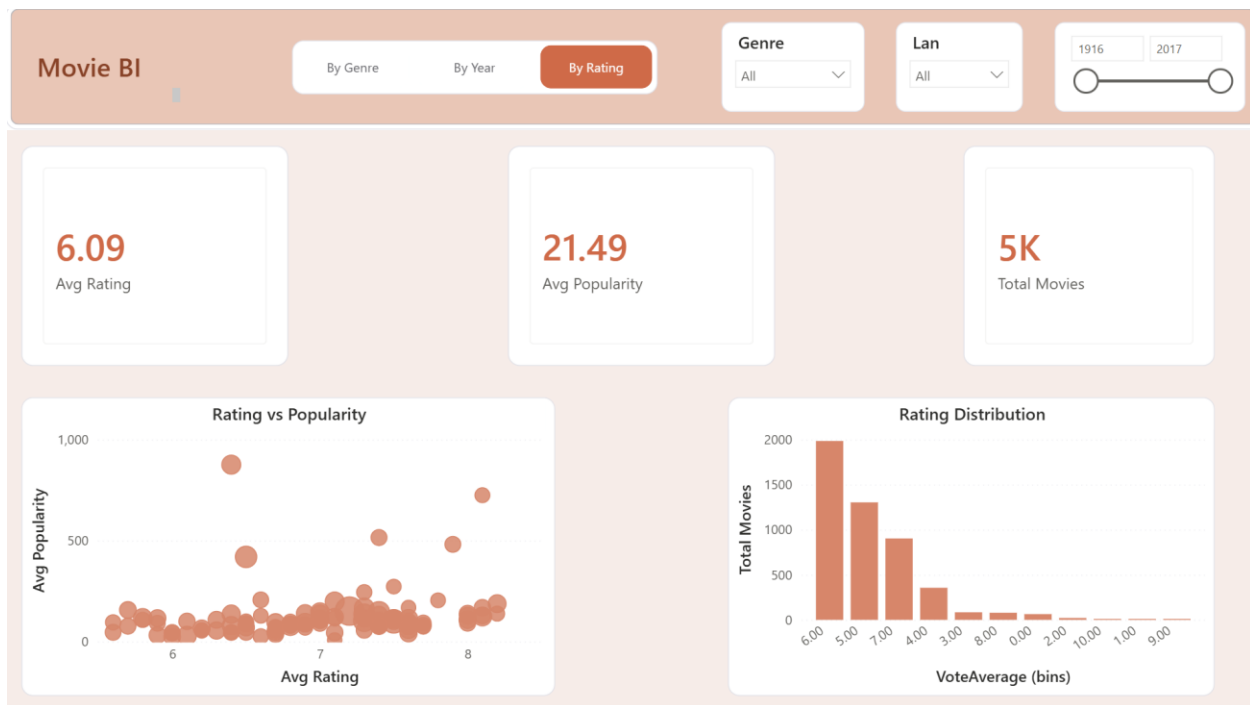
– Trend over years:

- Revenue and profit are near zero before ~1980, then grow exponentially, peaking around 2010 - 2013.
- Revenue increasingly outpaces budget over time, widening the profit gap - demonstrating that the industry scales profitably.



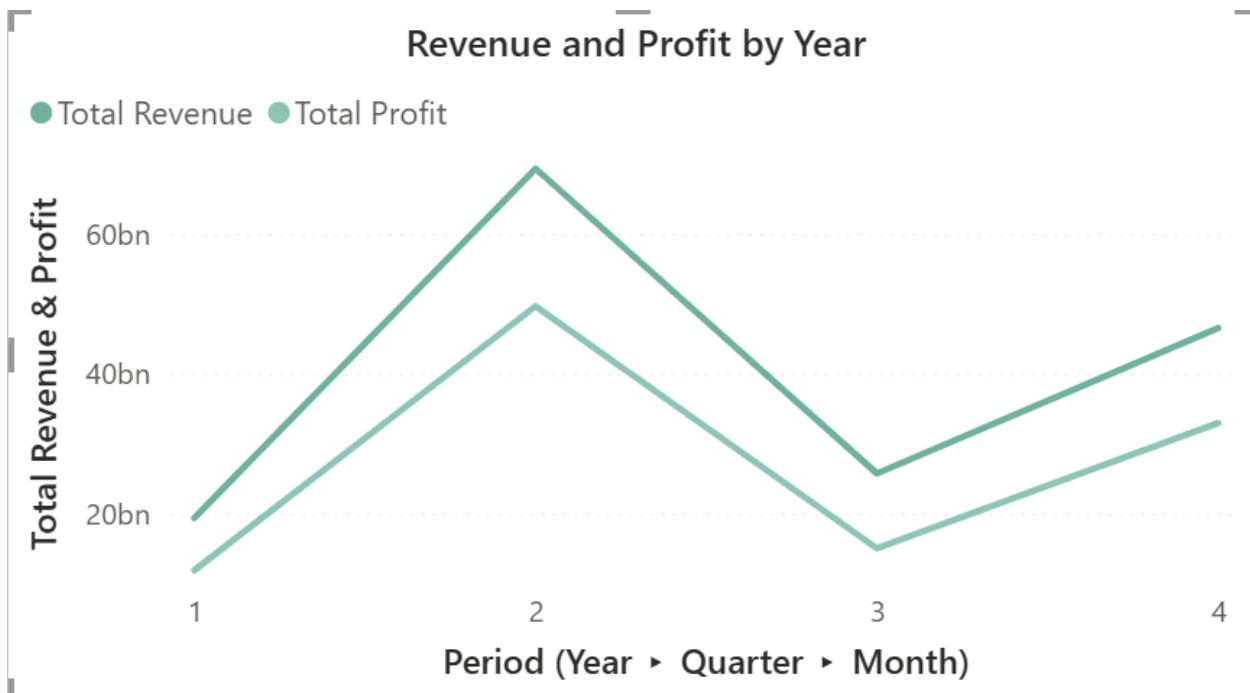
– Rating & popularity:

- Ratings cluster at 5 - 7 (mode 6, average 6.09) with few extremes - most films are of mid- tier quality.
- There is no clear correlation between rating and popularity; popularity is driven by a few cultural hits rather than by high scores.
- High revenue films sit at decent ratings (6.5 - 8) but rarely at the absolute top.

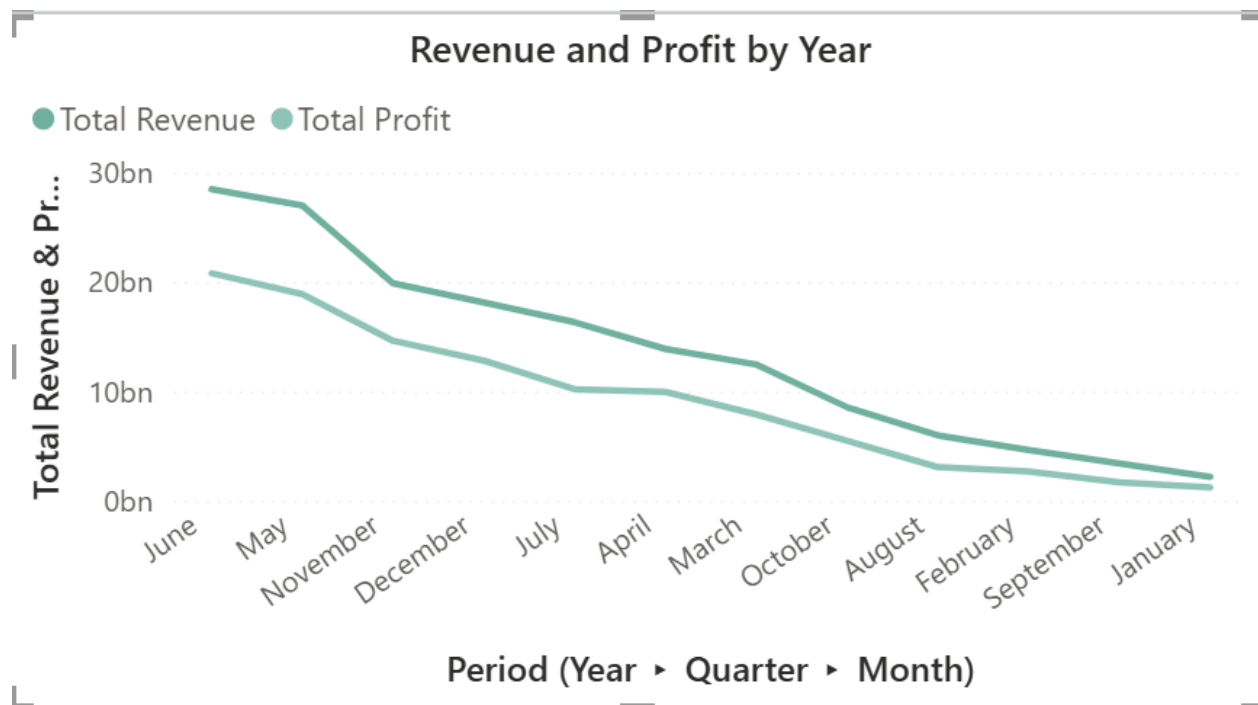


– Seasonality & release timing:

- Analyzing by quarter, Q2 generates the highest revenue (~\$135bn), followed by Q4 (~\$118bn). Q3 experiences a slight dip, while Q1 remains the lowest-performing quarter (~\$57bn).



- Breaking down by month, the top-earning periods are June, December, May, November, and July. This perfectly aligns with the classic blockbuster summer window (May - July) and the year-end holiday season (November - December). Conversely, January and February represent the weakest “dump months” of the year.



- Profit closely tracks revenue trends across all periods. Furthermore, the profit margin is notably higher during peak windows (for example: ~70% in June compared to ~55% in January). This indicates that prime release windows are not only larger in total volume but also more economically efficient.

→ Release timing is a critical factor for financial success. Studios and distributors should strategically schedule major film launches around the high-earning windows of June or November - December to maximize box office performance and ROI.

2. Trend evaluation and recommendations

– Trend evaluation:

- Revenue and profit are near zero before ~1980, accelerate sharply, and peak around 2010 - 2013. The dataset has matured into a high-revenue era (total ~\$395bn revenue, ~\$256bn profit).
- Revenue increasingly outpaces budget over time (budget ~\$140bn vs. revenue ~\$395bn), giving an aggregate margin of ~65% and ROI of ~1.8x on budget. Profitability scales, but it is concentrated in a few blockbusters rather than spread evenly.
- Drama, Comedy, and Thriller dominate by the number of films, yet Adventure and Action generate the most revenue and profit. This indicates that value is concentrated in a few high-budget genres, not necessarily the most-produced ones.
- Revenue peaks in summer (May - July, highest in June) and the year-end holidays (November - December); January and February are the weakest months. Profit margins are also slightly higher in these peak windows.
- Ratings heavily cluster at 5 - 7 (average 6.09) with no clear link between rating and popularity or revenue. Commercial success is driven by a handful of cultural hits, not purely by critical scores.

– Recommendations:

- Schedule high-budget tentpoles for June or November - December to maximize revenue, and aggressively avoid launching them during the January - February dump months.
- Adventure and Action generate the most revenue and profit despite having far fewer titles - meaning their revenue per film is much higher.

- Invest selectively in a few Adventure/Action tentpoles to drive massive revenue, while utilizing lower-budget Drama/Comedy/Horror films for volume and risk diversification. Constantly judge genres by per-film ROI, not by total count.
- Since revenue historically grows faster than the budget, maintain strict cost control. Verify low-budget/high-ROI hypotheses (e.g., the Horror genre) by actively comparing profit against budget per genre.
- Because critical ratings do not strongly correlate with popularity or revenue, studios must invest heavily in marketing and audience buzz rather than assuming high-quality films will sell themselves.

CHAPTER 7: CONCLUSION

1. Results

The team successfully built an end-to-end BI pipeline covering three stages:

- ETL (Python/pandas): Extracted, cleansed, and type-cast raw data. Parsed JSON-like genres into a relational format and handled placeholder zeros in Budget/Revenue (converting to NULL). Output: 5 clean CSV files.
- Data Warehouse (SQL Server): Implemented a Star Schema with a bridge table using TRY_CONVERT for safe loading. Total rows loaded: DimDate (~3,280), DimGenre (20), DimMovieInfo (~4,803), FactMovies (~4,803), and BridgeMovieGenre (~12,160).
- Dashboard (Power BI): Delivered an interactive report with 3 pages (By Genre, By Year, By Rating) using custom DAX measures and shared slicers to answer core analytical questions.

Overall, the system fulfills all functional requirements of the assignment and demonstrates a working, reproducible data-driven analytics workflow from raw CSV files to interactive insights.

2. Limitations and future development

- Limitations:
 - The dataset is a snapshot (up to ~2017), not reflecting the current market.
 - ~30% of movies lack budget/revenue data (set to NULL), restricting financial analysis to ~3,300 movies.
 - ETL, data loading, and Power BI refreshes are executed manually without automation.

– Future development:

- Build DimCast and DimCrew from the credits file to analyze actor/director impact.
- Use tools like SSIS or Apache Airflow for incremental loading and scheduled refreshes.
- Connect to the TMDB API to fetch real-time, up-to-date movie data.
- Layer machine learning models on the data warehouse to predict revenue or ratings.

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